Standard College Practical File / Record Format

Course: Computer Science / Engineering Laboratory | Student Name: [Name] | Roll No: [Roll No] | Semester: [Sem]

INDEX / LIST OF EXPERIMENTS

1. Implementation of Linear Search and Binary Search Algorithms
2. Implementation of Singly Linked List with Insertion and Deletion
3. Stack Implementation using Arrays with Push and Pop Operations
4. Queue Implementation using Circular Arrays
5. Implementation of QuickSort and MergeSort with Complexity Analysis

EXPERIMENT STRUCTURE GUIDE

Each laboratory experiment entry must follow this rigorous uniform structure:

1. Aim / Objective of the Experiment
2. Prerequisites / Theoretical Background
3. Hardware & Software Requirements
4. Algorithm / Flowchart
5. Documented Source Code / Circuit Diagram
6. Sample Input & Output Verification
7. Complexity Analysis / Viva Questions

SAMPLE PRACTICAL ENTRY: EXPERIMENT 01

AIM: Write a program to implement Binary Search on a sorted array of integers.

THEORY: Binary search operates on the divide-and-conquer paradigm with O(log n) time complexity.

ALGORITHM: Initialize low = 0, high = n - 1. While low <= high, compute mid = (low + high) / 2...

FACULTY EVALUATION CRITERIA

- Timely practical submission and index verification (5 Marks)
- Code correctness and adherence to coding guidelines (10 Marks)
- Viva Voce performance and concept clarity (10 Marks)
- Total Laboratory Assessment Weightage: 25 Marks